Chained Operations: (a + b) × (c - d) on the Stack

Numerical & Algebraic Expressions • Standard: CCSS.MATH.6.EE.A.2

BOOK STORY OPENER: Why 24 Is the Champion Packing Number, The Typesetter's Bargain & Dijkstra's 1961 Railway Siding

In the 1500s, typesetters invented parentheses only to squash 2D fraction bars onto flat lines of lead type—then in 1924 Jan Łukasiewicz proved that placing operators after their numbers eliminates every parenthesis, and in 1961 Edsger Dijkstra modeled the 4-register stack as a railway shunting yard! Alongside the 24-Tile Array Studio (where 24 packs into 1×24, 2×12, 3×8, and 4×6 rectangles), students evaluate nested expressions from the inside out.

Companion Book: From Blocks to the Stack (SC-6 Ch. 4: Why 24 Is the Champion Packing Number) & Thinking in the Stack (SC-32 Ch. 2: Dijkstra's Railway Siding)

1. CONCRETE MANIPULATIVE SETUP

Manipulative Kit: Counting & Array Tiles + 4-Register Stack

Arrange Counting & Array tiles to model the two sub-expressions: (6 + 1) = 7 rows by (9 - 5) = 4 columns.

2. STUDENT PREDICTION (ASK BEFORE PRESSING KEYS)

Left sub-expression: 6 + 1 = 7. Right sub-expression: 9 - 5 = 4. Combined 7 × 4 array: 7 × 4 = 28.

3. CALCULATOR ACTION (SC-6 KEYCAPS & STACK TRACE)

Keystroke Flow: [6] [ENTER] [1] [+] [9] [ENTER] [5] [-] [×]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
6	6	Value a (6) placed into Register X
ENTER	6	ENTER lifts 6 into Register Y
1	1	Value b (1) placed into Register X
+	7	Left operator (+): Combines Y(6) + X(1) = 7 in X
9	9	Value c (9) entered into X; 7 lifted to Y
ENTER	9	ENTER lifts 9 into Z and 7 into T
5	5	Value d (5) placed into Register X
-	4	Right operator (-): Combines Z/Y(9) - X(5) = 4 in X; 7 drops to Y
×	28	Root operator (×): Combines Y(7) × X(4) = 28 in X!

4. MATHEMATICAL EXPLANATION & RPN ADVANTAGE

Parentheses exist in written algebra only to compensate for linear notation. On the 4-line stack, precedence is intrinsic.

Core Investigation Problem:

Evaluate (8 + 2) × (14 - 9) by solving the inner quantities first, then checking on the stack.

Deep Stack Evaluation & Register Limits

Order of Operations & Stack Evaluation • Standard: CCSS.MATH.6.EE.A.2

1. FOLLOW-UP EXTENSION SETUP & PREDICTION

Setup: Track the 4 stack registers (T, Z, Y, X) for (12 - 4) / (2 × (1 + 1)). Prediction: Numerator: 12 - 4 = 8. Denominator: 2 × 2 = 4. Result: 8 / 4 = 2.

2. EXTENSION KEYSTROKE SEQUENCE & STACK TRACE

[1] [2] [ENTER] [4] [-] [2] [ENTER] [1] [ENTER] [1] [+] [×] [÷]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
[1]	1	First digit of 12
[2]	12	12 in register X
[ENTER]	12	12 lifted to Y
[4]	4	4 in register X
[-]	8	Numerator 12 - 4 = 8 held in X
[2]	2	Factor 2 enters X; 8 lifts to Y
[ENTER]	2	2 lifted to Y; 8 lifts to Z
[1]	1	Inner 1 enters X
[ENTER]	1	1 lifted to Y; 2 in Z; 8 in T
[1]	1	Second 1 enters X
[+]	2	Innermost sum: 1 + 1 = 2 in X
[×]	4	Denominator combined: 2 × 2 = 4 in X; 8 in Y
[÷]	2	8 / 4 = 2!

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Evaluating expressions strictly left-to-right (computing 10 + 5 × 2 as 15 × 2 = 30) instead of respecting operator structure.

Hands-On Block Remediation: Build the expression tree with Array Tiles: push 10 to Register Y ([10] [ENTER]), evaluate the 5 × 2 array in X ([5] [ENTER] [2] [×] -> 10), then press [+] -> 20.

Misconception 2: Losing track of intermediate results when evaluating two parenthesized branches like (8 + 2) × (9 − 4).

Hands-On Block Remediation: Show how Register Y holds the first branch (8 + 2 = 10) automatically while the second branch (9 − 4 = 5) is computed in X, ready for [×] = 50.

4. EXTENSION MATHEMATICAL INSIGHT

The 4-register stack can evaluate nested expressions up to depth 4 without spilling to storage registers.

Postfix Evaluation & Chained Operations: The 4-Register Stack

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: POSTFIX EVALUATION & STACK REGISTER RULES

- In postfix evaluation, operands are entered first, followed immediately by operators.
- Evaluation order matches mathematical precedence: evaluate inner sub-expressions first.
- RPN keystroke sequence evaluates complex expressions naturally without parenthesis keys.

Key Sequence: `6` `ENTER` `1` `+` `9` `ENTER` `5` `-` `×`

Problem 1: Evaluate (4 + 3) × (10 - 6) and check on the stack.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-6

Problem 2: Solve (20 - 4) / (2 × 4) using inner-first evaluation.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-6

Problem 3: Evaluate (18 - 6) / (1 + 2) on the 4-level stack.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-6

Problem 4: Evaluate 5 + (3 × 4) by computing the inner product first.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-6

Unit 07: Postfix Evaluation & Chained Operations: The 4-Register Stack — Answer Key

Complete solutions and SC-6 verification keystrokes for Grades 5–6

1. CORE & EXTENSION INVESTIGATION SOLUTIONS

Core Investigation: Evaluate (8 + 2) × (14 - 9) by solving the inner quantities first, then checking on the stack.

Core Solution: 50. Keystrokes: [8] [ENTER] [2] [+] [1][4] [ENTER] [9] [-] [×]

Core Check Keys: 8 ENTER 2 + 1 4 ENTER 9 - ×

Extension Challenge: Evaluate (15 - 5) / (3 + 2) by computing the numerator and denominator, then verifying on the stack.

Extension Solution: 2. Keystrokes: [1][5] [ENTER] [5] [-] [3] [ENTER] [2] [+] [÷]

Ext. Check Keys: 1 5 ENTER 5 - 3 ENTER 2 + ÷

2. STUDENT LAB WORKSHEET SOLUTIONS (PROBLEMS 1–4)

Problem 1: Evaluate (4 + 3) × (10 - 6) and check on the stack.

Expected Answer: 28

SC-6 Verification Keys: 4 ENTER 3 + 1 0 ENTER 6 - ×

Problem 2: Solve (20 - 4) / (2 × 4) using inner-first evaluation.

Expected Answer: 2

SC-6 Verification Keys: 2 0 ENTER 4 - 2 ENTER 4 × ÷

Problem 3: Evaluate (18 - 6) / (1 + 2) on the 4-level stack.

Expected Answer: 4

SC-6 Verification Keys: 1 8 ENTER 6 - 1 ENTER 2 + ÷

Problem 4: Evaluate 5 + (3 × 4) by computing the inner product first.

Expected Answer: 17

SC-6 Verification Keys: 3 ENTER 4 × 5 +

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.